

HR Analytics: Insights and Recommendations.

1. Attrition Overview

- The overall employee attrition rate is 16.12% — 237 out of 1,470 employees left the company.
- Research & Development records the highest number of employees leaving in absolute terms (largely because it is the company's largest department by headcount), but Sales has the highest attrition rate by department at 20.63%–21%.
- At the job-role level, Sales Representative has the highest attrition rate (39.76%), followed by Laboratory Technician (23.94%) and Human Resources role (23.08%). In absolute counts, Laboratory Technician (62), Sales Executive (57), and Research Scientist (47) lost the most employees.
- The average age of employees who left is 34 — attrition is concentrated among early/mid-career employees, not those near retirement.
- Employees under 30 account for over half of all attrition (52.83%) — more than double the 30–40 group (26.42%) and the over-40 group (20.75%).

2. Overtime & Attrition

- Employees who work overtime have an attrition rate of 30.53% — almost three times higher than those who do not work overtime, making overtime the single strongest behavioural predictor of attrition in this dataset.
- 46.41% of all employees who left were working overtime, despite overtime employees making up only 28.3% of the total workforce.
- Overtime employees are paid only slightly more than non-overtime employees — the gap is not significant, meaning overtime pay does not appear to compensate adequately for the added workload.
- Work-life balance and job satisfaction scores are nearly identical for overtime vs non-overtime employees (2.7–2.8 on both metrics), suggesting overtime's effect on attrition operates through factors beyond just these two scores — likely burnout and uncompensated effort.

3. Salary & Compensation

- Employees who left earned significantly less on average than those who stayed (\$4.8K vs \$6.8K — a ~29% gap). Statistically confirmed via two-sample t-test ($p < 0.05$): this gap is not due to chance.
- Monthly income varies significantly across departments — confirmed via one-way ANOVA ($p < 0.05$). Salary is genuinely influenced by department.
- Manager (\$17K) and Research Director (\$16K) earn far more than other roles. Research Scientist, Laboratory Technician, and Sales Representative sit at the bottom (~\$3K) — the same roles that record the highest attrition rates.
- Monthly income distribution across the company is heavily right-skewed — most employees earn between \$2,000 and \$5,000/month, with a long tail toward \$20K pulling the company-wide average up to \$6.5K.
- Employees who left cluster in the low-income, low-experience zone and fall below the company's typical pay-for-experience trend line.
- Average salary is broadly similar across all job satisfaction levels — confirming salary is not the only factor shaping satisfaction; work-life balance, overtime, and environment also contribute.
- Employees rated performance '4' receive roughly 21–22% salary hikes across all departments, versus ~14% for those rated '3'. HR has the slightly lowest hike in both band.
- Employees who stay longer generally earn higher salaries, though salary growth is not perfectly consistent across individuals.

4. Satisfaction & Engagement

- Average Environment satisfaction (2.7), average job satisfaction (2.7), and average work-life balance (2.8) all sit near the midpoint of the scale — broad, moderate dissatisfaction rather than one acute problem.
- Attrition rate falls consistently as job satisfaction rises — employees with higher job satisfaction scores are less likely to leave.
- Attrition rate by environment satisfaction follows an inverse pattern: lowest satisfaction score (1) → 25.35% attrition; highest score (4) → 13.45% attrition.
- A clear inverse relationship exists at department level: Sales and HR (lowest average environment satisfaction, 2.68) have the highest attrition (21% and 19%); R&D (highest average satisfaction, 2.74) has the lowest attrition (14%).
- Employees with the lowest work-life balance score (1) have an attrition rate of ~30% — nearly double the rates at scores 2 and 3. Attrition ticks back up slightly at score 4.
- Average work-life balance scores are very close between employees who left and those who stayed, indicating work-life balance alone does not explain attrition — it is concentrated among the lowest-scoring subgroup.
- Sales has the highest average job satisfaction score (2.8) yet also the highest departmental attrition rate — confirming job satisfaction alone does not explain attrition in Sales.
- Most employees report being satisfied: 459 rated job satisfaction as 4 (highest) and 422 rated it 3.
- Only 5 employees fall into the 'high risk' category — lowest possible scores on job satisfaction, environment satisfaction, and work-life balance simultaneously — 4 in R&D and 1 in Sales.

5. Demographics & Diversity

- The workforce is 60% male (882) and 40% female (588) — male employees outnumber female employees across all departments.
- Single employees have an attrition rate of 25.53%–26% — roughly 2–2.5x that of married (12%) and divorced (10%) employees.

6. Career Growth & Tenure

- Sales employees face a double bottleneck: the longest average time in current role (4.49 years) and the longest average wait for a promotion (2.4 years) — consistent with Sales having the highest departmental attrition rate.
- Managers and Research Directors wait the longest for promotion due to structural bottlenecks at leadership levels; entry-level Sales and HR roles see the fastest career movement.
- Promotion delay of 3+ years alone does not significantly increase attrition — attrition rates are fairly similar between the two promotion-wait groups.
- Average employee tenure is similar across departments (Sales: 7.3 years, HR: 7.2 years, R&D: 6.9 years). However, Sales still has the highest attrition because it has a mix of long-serving employees and entry-level roles, such as Sales Representatives, who tend to leave much sooner (average tenure: 2.9 years).
- Manager (14.4 years) and Research Director (10.9 years) have by far the longest average company tenure, while Sales Representative (2.9 years) has the shortest.
- Employees who have worked at 5 or more previous companies show a notably higher attrition rate, peaking at 25.40% — prior job-hopping behaviour is a strong predictor of future turnover.
- An average of 2.7 previous companies worked indicates most hires arrive with prior industry experience.
- Average training time is broadly similar across departments (Sales 2.85, R&D 2.79, HR 2.56) — training investment does not appear to be a strong differentiator of department-level outcomes.

7. Correlation Patterns

- Attrition and workplace behaviour are driven by a combination of factors rather than any single cause — most variables show only weak individual relationships with attrition.
- Substantial positive relationships exist between experience-related variables: Total Working Years & Monthly Income (0.77), Years at Company & Years with Current Manager (0.77), Years at Company & Years in Current Role (0.76), Years in Current Role & Years with Current Manager (0.71).
- Moderate relationships found between Monthly Income & Years at Company (0.51) and Total Working Years & Years at Company (0.63) — overall work experience is a major driver of salary growth.

8. Workforce Structure

- Research & Development is the largest department (dominated by Research Scientist and Laboratory Technician), followed by Sales (anchored by Sales Executive); Human Resources is the smallest.
- 'Travel Rarely' dominates across all departments; Sales shows a comparatively higher share of frequent travellers, consistent with its client-facing nature.

- Most employees fall in the lower-to-mid income range — the company has a large base of mid-level employees with only a small number of very high earners.

9. Recommendations

4. Combined Business Recommendations

1. Optimize Overtime Management

- Reduce excessive overtime by balancing workloads more effectively.
- Review overtime compensation and employee support, particularly in Sales, Laboratory Technician, and Research Scientist roles.

2. Reassess Compensation Strategy

- Conduct a salary review for roles with high turnover and lower pay levels.
- Ensure compensation remains competitive and aligned with employee responsibilities and market standards.

3. Invest in Early-Career Retention

- Develop targeted retention programs for younger and entry-level employees.
- Strengthen onboarding, mentoring, and career development initiatives to improve long-term engagement.

4. Accelerate Career Progression in Sales

- Review promotion processes and career pathways within the Sales department.
- Create clearer advancement opportunities to encourage employee growth and retention.

5. Enhance the Employee Experience

- Implement initiatives aimed at improving the workplace environment, particularly in Sales and HR.
- Focus on employee engagement, team culture, and manager support.

6. Strengthen Work-Life Balance Support

- Identify employees experiencing work-life balance challenges and provide targeted assistance.
- Encourage managers to proactively address workload and wellbeing concerns.

7. Strengthen Retention Efforts for High-Risk Employees

- Provide additional onboarding and engagement support for employees with a history of frequent job changes.

- Establish regular check-ins during the early stages of employment.

8. Implement Proactive Employee Check-Ins

- Conduct regular one-on-one discussions with employees showing signs of disengagement.
- Address concerns early before they develop into retention risks.

9. Establish an Attrition Early-Warning System

- Regularly monitor job satisfaction and workplace environment metrics.
- Use survey results and employee feedback to identify risks early and take corrective action.